Evaluating Agent Swarms: Token Budgets, Active Workers, and Baseline Choice

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Abstract

Evaluating an agent swarm requires distinguishing the effects of increasing the total token budget, dividing a fixed ceiling, and changing coordination. We examine these comparisons in 8,400 controlled trials on fresh arithmetic and graph-colouring tasks. At 4,096 completion tokens per worker, increasing independent width from one to four improves accuracy by 16.0 percentage points on both families while the total ceiling quadruples. Splitting a fixed 4,096-token completion ceiling instead lowers graph accuracy from 56.0% to 18.5%; the arithmetic change is inconclusive. The primary comparisons do not establish an accuracy advantage over a sequential control matched on scheduled calls and nominal token ceiling. Ring communication provides no consistent gain. Eleven complementary scaling studies add 20,376 held-out trials across seven task families, including a fresh-task sixteen-worker confirmation that does not establish superiority over fixed-four execution. Candidate and request audits show that finding a valid answer after an initial failure depends on the token allowance. They also reveal incomplete worker activation and 97 controlled-study attempts with unknown usage. The findings show how resource controls, output selection, and execution records qualify claims about agent scaling. They support evaluating specific allocation policies; equal realized compute advantages and general scaling laws remain unestablished from this work.

Keywords: Agent evaluation; Multi-agent systems; Test-time compute; Token budgets; Resource allocation; Coordination; Reliability; Reproducibility

1 Study and Findings

This ongoing research programme studies allocation, participation, and selection. Controlled comparisons use widths $N = 1, 2, 4$ (Figure 2). Complementary programming and ZebraLogic grids test $N = 1, 2, 4, 8, 16$; smaller diagnostic grids compare one and four. Tasks cover programming, logical reasoning, arithmetic, graph colouring, and planning (Zhuo et al., 2025; Liu et al., 2023; Lin et al., 2025; Kazemi et al., 2025; Valmeekam et al., 2023). Paired bootstrap intervals retain whole-task clusters and study-specific strata.

Figure 1 summarizes the complementary results. All 148 wider cells use more recorded tokens: accuracy is lower in 75, equal in 24, and higher in 49. At sixteen requested workers, BigCodeBench Muse's activation rises from 7.3% to 45.3% to 93.5% across ceilings of 32,768, 65,536, and 131,072. The transferred ZebraLogic $4/8/16$ policy gains $+0.92$ pp over fixed four (95% interval $[-0.58, 2.50]$), establishing neither superiority nor equivalence. The controlled fixed-total arithmetic estimate is -24.0 pp; the comparison remains inconclusive.

The wider grids reveal failures beyond four workers. For independent BigCodeBench Muse at $B = 32{,}768$, accuracy is 36/64 at $N = 8$, 14/64 at $N = 16$, and 40/64 at $N = 1$. Programming contributes 96 wider cells, 81 flat or worse, so the overall configuration count is not benchmark-balanced.

PlanBench's independent four-worker effect reverses from -95.0 pp ($[-98.0, -91.5]$) to $+10.5$ pp ($[6.0, 15.5]$) as its ceiling doubles; the higher-ceiling gain uses 4.06 times the control's retained debit. BBEH ring effects similarly change from -6.5 to $+4.5$ pp. Countdown's independent gains of 9.5–17.0 pp use 2.77–3.85 times the recorded tokens; ring revision instead reduces its pooled accuracy by 5.0 pp ($[-8.5, -1.75]$). These results separate search gains from the tested communication intervention.

Equal ceilings do not establish equal realized compute. Retained charges are incomplete ledgers, including 97 controlled-study attempts with unknown usage. Further work concerns resource-matched controls and policy transfer; general scaling laws remain unestablished. Evidence and protocols: https://github.com/PluvioXO/agent-swarms-reproducibility.

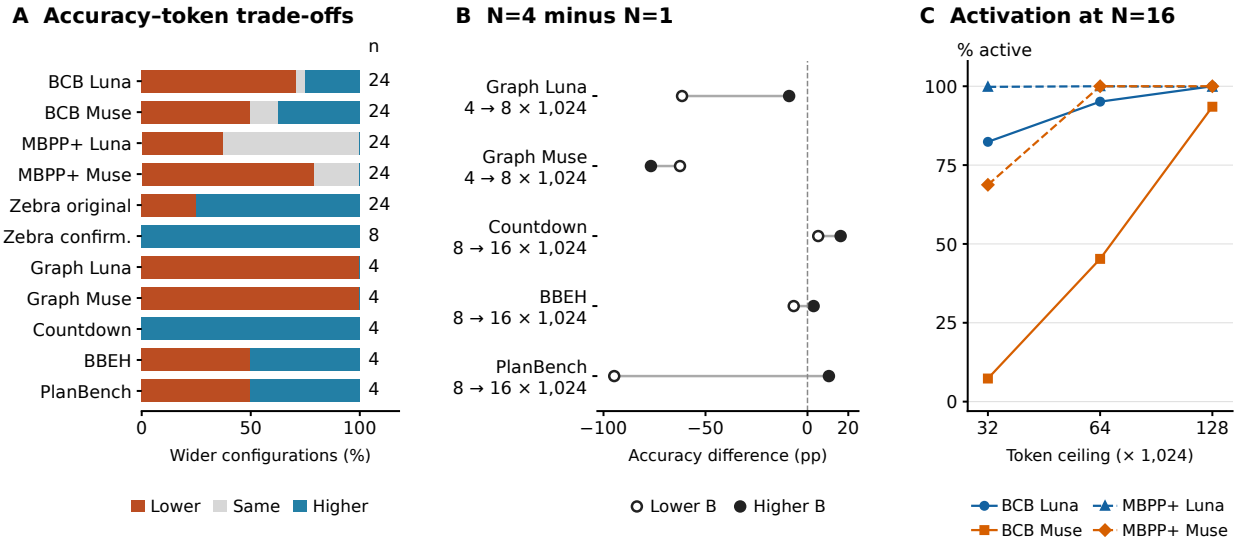


Figure 1 | Complementary results. (A) Accuracy at widths 2, 4, 8, and 16 where tested, versus matched-ceiling $N = 1$; all 148 wider cells use more recorded tokens. (B) Four-worker accuracy effects at two ceilings, pooled over revision conditions. (C) Activation at sixteen requested workers. Configurations share tasks and controls.

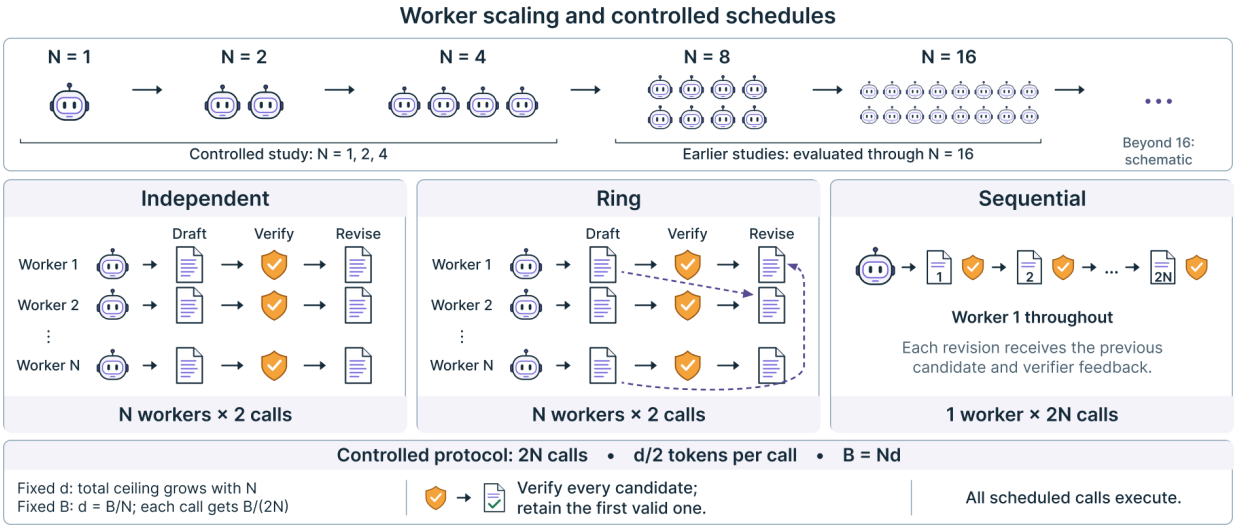


Figure 2 | Worker scaling and controlled schedules. N denotes parallel width, d the per-worker completion allowance, and $B = Nd$ the total completion ceiling. All controlled schedules execute $2N$ calls, each allowed $d/2$ completion tokens. The controlled study tests $N = 1, 2, 4$; earlier studies extend through $N = 16$. Widths beyond sixteen are schematic.

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